Pythagorean Triples

Typesetting Club

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1 Right Triangles with Whole Sides

A *Pythagorean triple* is a list of three positive integers (a, b, c) with $a^2 + b^2 = c^2$. By the theorem of Pythagoras, such numbers are the side lengths of a right triangle, as in Figure 1. Table 1 lists the five triples with the smallest hypotenuse c.

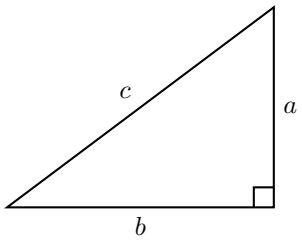


Figure 1: A right triangle with legs a, b and hypotenuse c

Table 1: The five Pythagorean triples with the smallest c

a	b	c	$a^2 + b^2$
3	4	5	25
6	8	10	100
5	12	13	169
9	12	15	225
8	15	17	289

2 Euclid's Formula

Most triples in Table 1 come from one formula, which was already known to Euclid.

Theorem 1 (Euclid)**.** *Let m and n be integers with $m > n > 0$. Then*

$$a = m^2 - n^2, \qquad b = 2mn, \qquad c = m^2 + n^2$$

form a Pythagorean triple.

Proof. Since $m > n > 0$, all three numbers are positive. Expanding gives

$$\begin{aligned} a^2 + b^2 &= (m^2 - n^2)^2 + (2mn)^2 \\ &= m^4 - 2m^2n^2 + n^4 + 4m^2n^2 \\ &= m^4 + 2m^2n^2 + n^4 \\ &= (m^2 + n^2)^2 = c^2. \end{aligned}$$

□

For example, $m = 2$, $n = 1$ gives $(3, 4, 5)$ and $m = 3$, $n = 2$ gives $(5, 12, 13)$. The triple $(9, 12, 15)$ does not come from Theorem 1 directly: it is three times $(3, 4, 5)$.

Verification Log

Piece	**What the AI got wrong**	**How I checked and fixed it**
P1	It loaded a package called amsproof, but no such package exists.	Compiling stopped with an error, so I deleted that line.
P2		
P3		
P4		
P5		